

Aykut Arslan, PhD

AI Research Engineer | Foundation Models, Agents, Post-Training, and Evaluation

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PROFILE

AI research engineer and applied mathematician working across foundation-model training, post-training, autonomous agents, frontier-model evaluation, and formal reasoning. I build reproducible experimental systems for long-horizon agent evaluation, training-data quality, automated verification, and rigorous failure analysis.

EXPERIENCE

Tundra Technical Solutions - Client: Meta

GenAI Knowledge Expert - Frontier AI Research | Aug 2025 - Present

- Evaluate realistic agentic environments and red-team benchmark validity across software engineering, science, finance, infrastructure, and other professional domains.
- Work on foundation-model training and post-training, controlled ablations, distributed PyTorch experimentation, and frontier-model evaluation.
- Build reproducible evaluation infrastructure for sandboxed execution, trace replay, automated verification, evaluator reliability, and reward-signal integrity.

Snorkel AI

Consultant - AI Evaluation & Data Quality | May 2024 - Aug 2025

- Mathematical, scientific, multimodal, and long-form reasoning data for AI training and evaluation.
- Source-grounded validation, expert calibration, evaluator QA, adversarial reasoning examples, and structured error taxonomies.

Humanity's Last Exam

Lead STEM Domain Expert - External Research Collaboration | Aug 2024 - Nov 2024

- Expert verification and multi-rater review for advanced STEM benchmark problems, with structured rubrics and adjudication criteria.

University of Southern California

Assistant Professor, Applied Mathematics and Statistics | Aug 2023 - May 2025

- Machine learning, deep learning, probability, statistics, numerical methods, and applied mathematics.
- Research in high-dimensional inference, robust statistics, optimization, and false-discovery control.

University of Southern California, Lv Lab

Research Assistant | Aug 2017 - Aug 2023

- Robust high-dimensional statistical inference and reproducible simulation research.

RESEARCH & ENGINEERING

Learning When to Intervene in Software Agents

Developing a framework for learning when an autonomous software agent should continue its current trajectory versus reassess its strategy, using verified downstream execution outcomes rather than subjective judgments of intermediate actions.

Use matched counterfactual continuations and measure complete held-out task success under fixed budgets.

Engineering: Built infrastructure for trajectory checkpointing, state restoration, execution-identity validation, authoritative outcome parsing, budget accounting, holdout protection, and reproducible policy evaluation.

Research principle: Evaluation is based on authoritative software execution outcomes rather than whether an intermediate action merely appears reasonable.

Benchmark Reliability at Scale

Audited a 109-file benchmark environment containing approximately 540k tokens, machine-checking 100 cited evidence passages, validating page/row/line citations, independently re-deriving quantitative results, and reproducing ground truth for 8 of 9 evaluated questions.

Focused on semantic failures that survive schema and formatting checks: source-of-truth contradictions, inconsistent rubrics and test suites, and scoring defects capable of silently corrupting evaluation or training signals.

Evaluator Reliability

Evaluate multi-stage LLM-judge systems against expert-grounded labels, measuring defect recall, false-clean behavior, localization accuracy, evaluator disagreement, severity agreement, and failure attribution.

SELECTED PUBLICATIONS

The Strict Threshold for Gaussian Ellipsoid Fitting

A. Arslan. Meta AI Research, 2026.

Tightness of the Cycle-Based Relaxation for Completed Length-Three Alpha-Cycles

A. Arslan. Meta AI Research, 2026. Complete, machine-checked Lean formalization.

Adaptive Decision Policies for Long-Horizon Research Agents with Recursive Policy Improvement

A. Arslan. Submitted to ICLR 2027.

TECHNICAL SKILLS

Python, PyTorch, Lean, SQL, Linux, Docker, Git, AWS Bedrock, LLM APIs, distributed PyTorch training, mixed precision, checkpointing, sandboxed execution, trace replay, automated verification, agent evaluation, LLM-as-a-judge, experimental design, high-dimensional statistics, optimization, formal verification.

EDUCATION

University of Southern California - Ph.D., Applied Mathematics (Statistics Concentration), 2017-2023; M.S., Statistics, 2021-2023.

Middle East Technical University - B.S., Mathematics.